



A new hybrid vision transformer model for medical image classifications: Attention mechanism deepened by HOG

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ABSTRACT

In medical image classification, factors such as limited sample size, unbalanced data distribution, and variable resolution can negatively impact the performance of deep learning-based models. In this study, we propose Feature Fusion-ViT, a hybrid classification model that combines the global attention mechanism of the Vision Transformer (ViT) architecture with Oriented Gradient Histogram (HOG)-based local structural feature extraction. In the proposed architecture, local features extracted by HOG are integrated with marker embeddings learned through concatenation by ViT before the classification head. By combining ViT's high-level global representations with HOG's edge and texture sensitivity, the model aims to improve classification performance on medical images under adverse conditions such as unbalanced data distribution and variable resolution.

The proposed method was evaluated on the CBIS-DDSM dataset, consisting of 12 subdatasets from the MedMNIST collection and high-resolution clinical mammograms. Experimental results demonstrate that FeatFusion-ViT performs highly and competitively not only with ViT-based models in terms of accuracy and AUC-ROC, but also with hybrid approaches from the literature, such as SIFT+CNN and HOG-LBP+CNN. Specifically, on the CBIS-DDSM dataset, the model achieved an average performance improvement of 2%–3% in sensitivity and specificity compared to previous studies. Furthermore, Grad-CAM analyses demonstrate that the model consistently focuses on clinically relevant regions within the image when making decisions, confirming that the proposed architecture has an interpretable and reliable decision mechanism. In conclusion, the FeatFusion-ViT architecture offers a classification approach that provides high accuracy and clinical applicability for both low-resolution and high-resolution medical images.

1. Introduction

Medical image classification is a fundamental component that directly affects the accuracy and reliability of modern digital diagnostic systems. These classification processes are widely used to increase the effectiveness of clinical decision support systems that doctors use to make vital decisions such as diagnosis, treatment planning, and patient management. However, medical images often present data structures that present various challenges. These challenges include low resolution, high intra-class variation, unclear distinctions between classes, low signal-to-noise ratio, and especially a limited number of labeled examples [1]. This makes it difficult to obtain meaningful representations to be learned from images and limits the generalization ability of deep learning models. Especially in small datasets, the tendency of the model to over-training poses a significant risk for systems requiring high precision in medical applications [2]. Convolutional Neural Networks (CNN) have long been preferred in the field of medical image analysis due to their success in extracting local features in images

and recognizing patterns among these features. However, CNNs are inadequate in modeling global context because they work on patterns in a limited area due to their structure [3]. This limitation restricts model performance, especially in medical images where there are small differences between classes and contextual cues are distinctive. Vision Transformer (ViT) architectures, which have emerged in recent years, divide the image into fixed-size patch pieces, put these patches in order, and process them through a multi-head attention mechanism together with positional coding. This structure enables ViT to effectively model relationships between distant regions in the image [3]. In this way, ViT-based models can exhibit high performance, especially in complex medical image classification tasks, and provide better generalization compared to CNN-based approaches [4]. However, the fact that ViT models contain a large number of parameters and require a large amount of labeled data during training poses a serious disadvantage in data-constrained areas such as medical imaging [5]. The lack of

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labeled data can cause ViT to fail to sufficiently train its attention mechanism and, therefore, fail to model inter-class distinctions correctly. In this context, a more careful model architecture design is required in medical image classification problems that include a limited number of examples and where inter-class distinctions are visually ambiguous. In such cases, modeling additional information sources is quite valuable to enrich the learning process and increase the generalization ability of the model. Combining manually extracted statistical or structural features with ViT-extracted representations, such as Histograms of Oriented Gradients, can enable the model to process information at both local and global levels. Such a hybrid structure can support the learning of more powerful representations in low-data conditions while preserving the global context learning advantage of ViT.

In this study, “FeatFusion-ViT” is proposed as a hybrid deep learning architecture for medical image classification tasks. This architecture integrates Vision Transformer (ViT), a new generation global attention-based model, with the Histogram of Oriented Gradients (HOG) method, a traditional hand-crafted feature extraction technique, bringing together both learned and statistical representations. The ViT architecture divides the input image into fixed-size patches, adds position information to these patches, and provides a strong global representation by modeling long-range dependencies with a multi-layer multi-head attention mechanism. However, large-scale datasets are usually required for these structures to be learned effectively. This leads to performance constraints in medical imaging areas where the number of samples is limited and the label distribution is unbalanced. In order to overcome this limitation, the low-dimensional structural and edge-based information obtained by HOG feature extraction in the “FeatFusion-ViT” structure is processed in parallel with the high-dimensional representations learned by ViT and combined in the classification phase. Thus, ViT’s global context learning ability and HOG’s local edge sensitivity are combined under a single decision mechanism. HOG features are obtained from grayscale input images, while ViT processes the RGB version of the same image in a patch-based manner. These two representations are combined before fully connected layers and contribute to the classification decision. This integration supports the generalization power of neural network-based feature representations with the interpretability and structural sensitivity of traditional methods.

The proposed method was systematically evaluated on 12 different subsets within the MedMNIST [6] data collection system. These subsets include binary classification, multi-class, and multi-labeling tasks. MedMNIST, with its diverse labeling structures, class numbers, unbalanced data distributions, and image types, provides a suitable environment for evaluating the generalizability and adaptability of the method. Additionally, the Curated Breast Imaging Subset of the Digital Database for Screening Mammography (CBIS-DDSM) [7] dataset was utilized to evaluate the model’s performance on high-resolution mammograms under real clinical conditions. This dataset consists of real mammographic images labeled by expert radiologists, allowing the proposed method to be evaluated for accuracy and robustness on clinical-level image data. Experimental results demonstrate significant improvements in accuracy and AUC-ROC values compared to ViT-based models alone, particularly when inter-class similarity is high or the number of samples is limited. Additionally, the proposed method is compared with hybrid approaches that combine hand-crafted and deep features, such as SIFT + CNN and HOG-LBP + CNN. These comparisons show that the proposed model achieves competitive results with these methods in terms of both accuracy and overall classification performance. The contributions of the study can be summarized as follows:

- The “FeatFusion-ViT” framework combines the global context representation of ViT with the local edge and texture information of HOG to provide a more comprehensive feature representation.
- The decision mechanism is strengthened by integrating the low-dimensional features extracted by HOG into the classification layer in parallel with the ViT outputs.

- The proposed method has been tested on multiple subsets of the MedMNIST dataset and high-resolution CBIS-DDSM mammograms, achieving high accuracy and AUC-ROC values for both different medical imaging tasks and clinical-level data.
- The proposed method achieved superior performance in terms of accuracy and AUC-ROC compared to classical ViT and CNN-based deep learning models, as well as hand-crafted deep hybrid approaches such as SIFT+CNN and HOG-LBP+CNN.

2. Related work

Many studies have been conducted to improve the classification accuracy of medical images. Due to the challenging nature of the MedMNIST dataset, many studies have been conducted on this dataset. Some of the studies are; Prakash et al. [8] proposed the use of Additive Angular Margin Loss (ArcFace) function instead of traditional softmax loss function to improve the performance of ResNet-18 [9] network in medical image classification tasks. They increased the performance of the model by tuning the margin (m) and scale (s) hyperparameters. Singh et al. [10] conducted a medical image classification experiment using purely quantum-based models. They designed a QML workflow that can be implemented on real hardware by integrating techniques such as dynamical decoupling, gate twirling, and M3 error mitigation to reduce the error effects in quantum hardware. Lee et al. [11] conducted a comparative analysis of various active learning strategies on the MedMNIST dataset, aiming to reduce the labeling cost in medical image classification tasks. In particular, they systematically evaluated how popular AL methods such as entropy sampling, BALD, MC-Dropout, and corset perform with different classifiers. They emphasized that the model selection strategy and class balance in the active learning loop play an important role in the success. Ali et al. [12] proposed a Neural Architecture Search (NAS) method based on evolutionary algorithms for 2D and 3D medical image classification tasks. Their proposed method aims to automatically discover the most suitable deep learning architecture across different data types and dimensional structures. They focused on developing more efficient and scalable classification systems through automated architecture discovery in medical image analysis. Johnson et al. [13] proposed a hybrid neural network architecture combining classical deep neural networks with quantum transformations to improve performance in image classification tasks. With their proposed model, they investigated the contribution of quantum transformations, especially in making the feature space more discrete and reconstructing the representations in a more meaningful way. Manzari et al. [14] proposed a novel Transformer architecture that integrates the dilated Neighborhood Attention mechanism and the Kolmogorov-Arnold Networks (KAN) structure in order to effectively capture both local and global context in medical image classification tasks. They aimed to improve the separation between classes by modeling larger spatial dependencies with less computational cost compared to classical self-attention mechanisms. With the KAN module, they did not contribute to the generalization performance of the model, especially in low data structures, thanks to its ability to represent nonlinear functions with high accuracy. Chen et al. [15] proposed a quantum-enhanced diffusion model for use in medical image datasets such as MedMNIST. In their study, they aimed to improve image generation quality by integrating parametric quantum circuits (PQC) into the bottleneck part of the classical U-Net architecture. Wu and Papiez [16] presented a systematic comparison of the MedMNIST dataset to comprehensively evaluate the effectiveness of large-scale baseline models in medical image classification. Different architectures, such as Vision Transformer (ViT), Swin Transformer, and ConvNeXt, were analyzed in terms of model size, number of parameters, and training strategies and tested for their suitability for clinical scenarios. Wilhelmi and Rusiecki [17] investigated the applicability of a simple CNN architecture instead of large-scale pre-trained models for medical image classification tasks. They showed that a simpler

CNN model can give competitive results compared to multi-layered and complex structures with the right architecture and hyperparameter settings. Haque et al. [18] proposed a novel hierarchical Transformer architecture, SwinMedNet, to perform high-accuracy classification for medical images across modalities. The proposed architecture combines a Swin Transformer-based local window-based attention mechanism with cross-modality learning to optimize both local and global information flow. Sajid et al. [19] proposed a hybrid method combining deep learning-based features (CNN) with hand-crafted features (HOG and LBP) to improve breast cancer classification performance from mammography images. In their study, they classified images as benign or malignant using the CBIS-DDSM dataset. For this purpose, they developed a compact 12-layer CNN architecture called cVGG. The deep features obtained from the cVGG model were combined with HOG and LBP features, and the resulting hybrid feature vector was compared with XGBoost, Random Forest, and KNN classification algorithms to evaluate its performance. Mannarsamy et al. [20] proposed a new hybrid method called SIFT-BCD to improve breast cancer detection from mammography images. This method aims to improve classification performance by combining SIFT and CNN-based feature extraction techniques. The study used the CBIS-DDSM dataset, and mammography images were classified as normal, benign, and malignant. The research results demonstrated that the integration of deep learning methods with classical image processing techniques provides high accuracy rates and low false positive/negative rates in early-stage breast cancer diagnosis. Sharma et al. [21] proposed a deep learning approach for breast cancer detection that includes preprocessing, segmentation, and classification stages on CBIS-DDSM and Breast Histopathology Images datasets. In their study, they applied a median filter, sharpening, and CLAHE to mammography images for contrast enhancement, followed by segmentation using Watershed and Canny edge methods. Athsayamani et al. [22] proposed a novel approach combining improved denoising filters with an attention mechanism and a residual connection CNN structure to improve classification performance by reducing noise in MRI images. They improved image quality with improved versions of classical filters and achieved more robust feature extraction with CNN architectures. Huang et al. [23] systematically examined self-supervised learning approaches used in medical image classification tasks to reduce the need for labels and enhance model generalization, and presented a comprehensive analysis evaluating the performance of these methods across different medical modalities. Kowsari et al. [24] proposed the HMIC approach, which combines CNN models operating at different levels by treating the medical image classification problem as a hierarchical structure rather than a simple multi-classification process. They achieved higher accuracy compared to traditional methods in high-level (EE-CD-Normal) and low-level (Marsh scores) classifications.

3. Materials and methods

Many studies on medical image datasets indicate that deep learning models greatly enhance medical image classification performance. In the literature, attempts have been made to increase the classification performance with methods such as Transformer architectures, quantum-assisted approaches, active learning strategies, and evolutionary architecture search, but these approaches are generally limited by high computational cost, large data requirements, or explainability problems. In this study, in order to preserve both computational efficiency and enhance class separation, the hand-crafted feature extraction technique HOG and the powerful global context modeling capability of ViT architecture are combined in a hybrid manner. The proposed approach aims to increase the classification accuracy by combining the low-level edge and texture information of HOG with the high-level representations of ViT in parallel.

Table 1

Details of datasets used.

Dataset	Classes	Number of data			Samples
		Train	Val	Test	
PathMNIST	9	89,996	10,004	7180	107,180
ChestMNIST	14	78,468	11,219	22,433	112,120
DermaMNIST	7	7007	1003	2005	10,015
OCTMNIST	4	97,477	10,832	1000	109,309
PneumoniaMNIST	2	4708	524	624	5856
RetinaMNIST	5	1080	120	400	1600
BreastMNIST	2	546	78	156	780
BloodMNIST	8	11,959	1712	3421	17,092
TissueMNIST	8	165,466	23,640	47,280	236,386
OrganAMNIST	11	34,581	6491	17,778	58,850
OrganCMNIST	11	13,000	2392	8268	23,660
OrganSMNIST	11	13,940	2452	8829	25,221
CBIS-DDSM	2	2545	365	737	3647

3.1. Dataset

In this study, two different datasets were used to evaluate the performance of the developed model on both low-resolution benchmark images and high-resolution clinical mammograms. MedMNIST [6] is a standardized dataset composed of small-sized images obtained from various medical imaging modalities (biopsy, dermoscopy, fundus, X-ray, etc.) created for benchmarking deep learning models. The dataset contains images resampled to 28×28 pixels and covers various medical imaging tasks such as chest X-rays, skin lesions, blood cells, and tissue samples. In this study, two-dimensional subsets of the MedMNIST dataset were used for classification tasks. To evaluate the model's performance on real clinical images, the CBIS-DDSM [7] dataset was also used. CBIS-DDSM is a subset of the Digital Database for Screening Mammography (DDSM) [25] database, selected and relabeled by expert radiologists. The dataset includes mass and microcalcification regions found in mammographic scans, and for each sample, annotations such as BI-RADS assessment categories, lesion type (benign/malignant), and location information are included. Using both datasets together allowed for a comparative examination of the model's accuracy and generalizability across different resolutions and imaging conditions. Basic information about the datasets is shown in Table 1.

Some images of the classes in the datasets are seen in Fig. 1.

3.2. Histogram of Oriented Gradients (HOG)

Histogram of Oriented Gradients [26] is an orientation-based feature extraction method used to represent object structure and shape information in images. First proposed for pedestrian detection, this method exhibits successful performance especially in identifying structural information such as edges, corners and contours. The HOG algorithm is based on dividing the image into local regions and statistically summarizing the gradient orientations in each region. Thus, both more robust representations against illumination changes are produced and geometric structures are preserved [27]. The HOG feature extraction process basically consists of four main stages:

Gradient Calculation : Gradient values in horizontal G_x and vertical G_y directions for each pixel of the image are usually calculated using derivative masks such as the Sobel filter, as seen in Eq. (1).

$$G_x = I(x+1, y) - I(x-1, y) \quad (1)$$

$$G_y = I(x, y+1) - I(x, y-1)$$

Here $I(x, y)$ represents the pixel intensity value at x, y location of the image. The gradient magnitude and direction for each pixel are calculated as in Eq. (2)

$$|G| = \sqrt{G_x^2 + G_y^2} \quad (2)$$

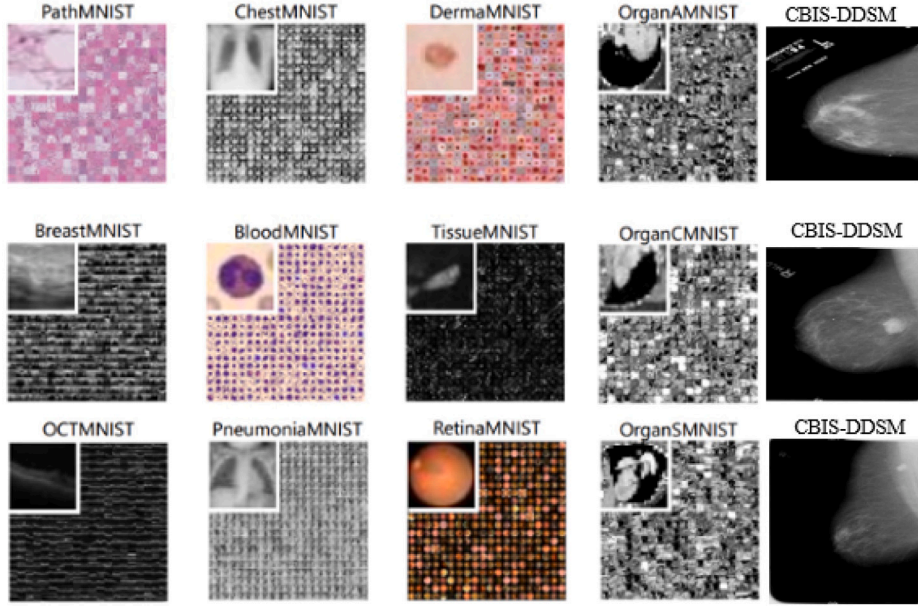


Fig. 1. Sample images from datasets.

$$\theta = \arctan\left(\frac{G_y}{G_x}\right)$$

The gradient direction θ is usually defined in the range 0° – 180° (signed) or 0° – 360° (unsigned).

Partitioning and Orientation Histogram : The image is usually divided into 8×8 pixel cells. A histogram is created by a weighted voting method corresponding to the gradient directions of pixels in each cell. Histograms are defined at equal angular intervals according to a certain number of directions B (e.g., 0° – 180° with 20° intervals for $B = 9$). The contribution of pixels falling in each direction range is calculated by weighting them with magnitude $|G|$.

Block Normalization: To reduce sensitivity to illumination and contrast changes, blocks of several cells (usually 2×2 cells) are merged and the histogram vector within this block is normalized. One of the most commonly used methods is $L2$ – *norm* normalization, seen in Eq. (3):

$$v_{norm} = \frac{v}{\sqrt{\|v\|_2^2 + \epsilon^2}} \quad (3)$$

Here v is the vector formed by the union of the histograms within the block and ϵ is a small constant for numerical stability ($\epsilon \approx 1e^{-6}$).

Creating Feature Vector: The normalized histograms obtained from all blocks are combined to obtain a fixed-size feature vector. This feature vector is made ready to be used as input in high-level tasks such as classification. The HOG feature map of an image from the Retinamnist dataset within Medmnist is shown in Fig. 2.

3.3. Vision Transformer (ViT)

Vision Transformer is an image data adaptation of the Transformer architecture, which was developed as an alternative to convolutional neural networks (CNN) in image classification tasks and is widely used in the field of natural language processing [28]. ViT can provide high classification performance, especially on large datasets, by modeling the global context using the self-attention mechanism instead of convolutional layers.

Splitting the Image into Patches: The input image is of size $x \in \mathbb{R}^{H \times W \times C}$, where H , W , and C represent the height, width, and number of channels, respectively. The image is divided into $N = \frac{HW}{P^2}$ 2D patches of size $P \times P$. These patches are flattened and converted into vectors as seen in Eq. (4).

$$x_p^i = \text{Flatten}(x_{(i)}) \in \mathbb{R}^{P^2 \cdot C}, i = 1, \dots, N \quad (4)$$

Linear Projection and Position Coding: Each patch vector is transformed into a fixed-size linear projection as in Eq. (5).

$$z_0^i = x_p^i \cdot E, E \in \mathbb{R}^{(P^2 \cdot C) \times D} \quad (5)$$

Here D represents the embedding size. Then, a location code p_i corresponding to each patch is added as shown in Eq. (6).

$$z_0^i = x_p^i \cdot E + p_i \quad (6)$$

Additionally, a special “class token” $z_0^{(cls)}$ is added at the beginning of the entire array, which is used for classification purposes.

Transformer Encoder Layers: ViT contains multi-layer encoder blocks, as in classical Transformer architectures. Each encoder block consists of Multi-Head Self-Attention (MHSA) and Feed-Forward Network (FFN) components [29]. The updated representations for an encoder layer are expressed as in Eq. (7).

$$z_l' = MSA(LN(z_{l-1})) + z_{l-1} \quad (7)$$

$$z_l = MLP(LN(z_l')) + z_l'$$

Here, LN represents the layer normalization and MLP represents the feedforward network consisting of two linear layers. Residual connections are used between the layers [30].

Classification Layer: The class token vector $z_L^{(cls)}$ obtained from the Transformer encoder is finally routed to the classification header via a linear layer as in Eq. (8).

$$y = \text{softmax}(W \cdot z_L^{(cls)} + b) \quad (8)$$

Here $W \in \mathbb{R}^{K \times D}$, K is the number of classes and D is the embedding dimension. ViT architecture can effectively model pixel relationships in distant regions by creating attention-based global representations instead of convolution [31]. However, it requires large-scale pre-training or transfer learning to achieve high performance on small datasets [32]. In this study, we propose a combination of a ViT-based architecture with traditional feature extraction methods, such as HOG, to evaluate both low-level edge information and high-level contextual representations together. The general diagram for ViT is shown in Fig. 3.

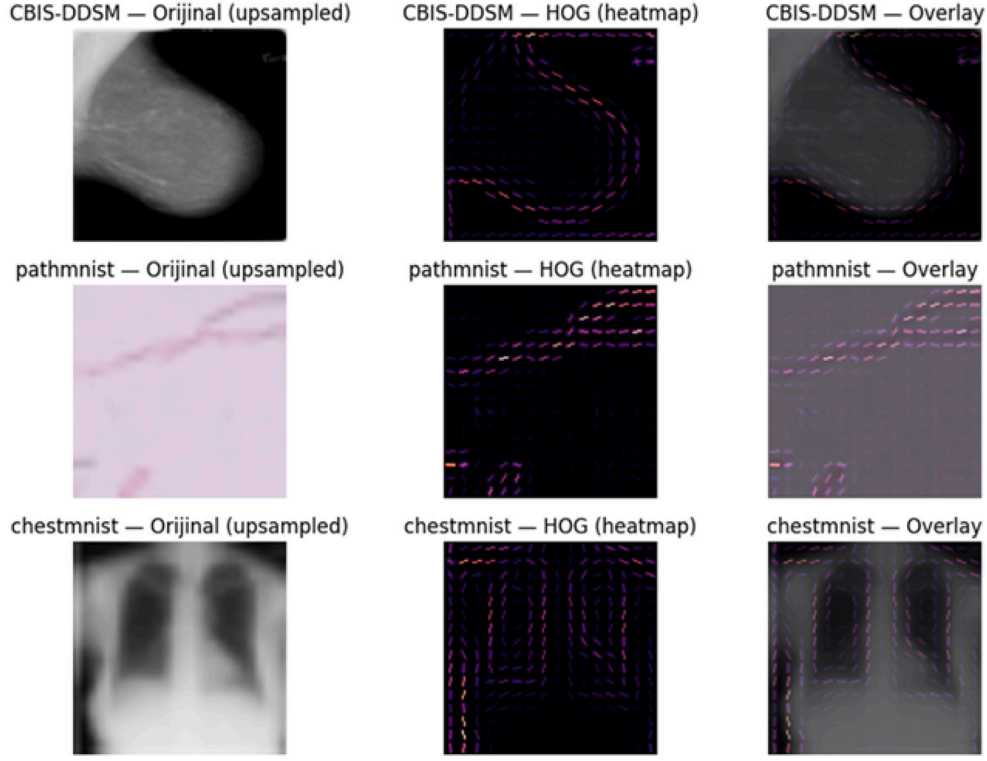


Fig. 2. Image of example HOG feature maps for the MedMNIST and CBIS-DDSM datasets.

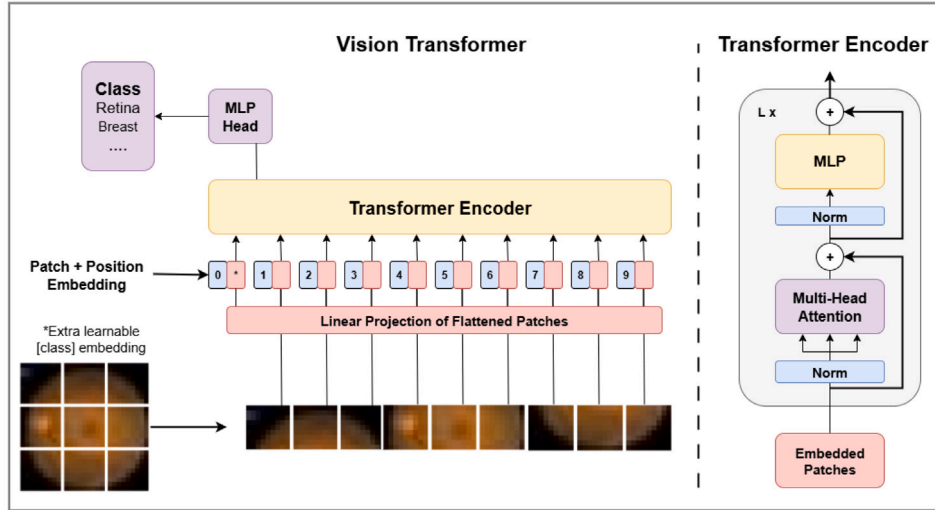


Fig. 3. The general diagram for ViT.

4. Proposed method: Feature Fusion Vision Transformer (*FeatFusion – ViT*)

The “FeatFusion-ViT” method proposed in this study is a novel hybrid classification architecture that combines local structural features with global context information in medical image classification tasks. This model aims to integrate the strengths of the traditional hand-crafted feature extraction, HOG, and the ViT architecture, simultaneously utilizing the advantages of both structures. In this way, the model combines both low-level (such as local contour, edge) and high-level (semantic context, object structure) information in the same feature space, thus improving the classification accuracy.

Input Image Preparation: Each input image $X \in \mathbb{R}^{H \times W}$ is normalized and directed to both ViT and HOG feature extraction paths. If

necessary, it is converted to grayscale. At this point, each image will be processed in two parallel paths. **Extraction of HOG Features:** The HOG algorithm represents the edge, texture and structural information in the image via directional gradient histograms. The image X is divided into cells and blocks, and the directional gradient information in each cell is converted into a histogram as in Eq. (9).

$$f_{HOG} = HOG(X; \theta_{HOG}) \in \mathbb{R}^{d_h} \quad (9)$$

Here;

- θ_{HOG} : It includes hyperparameters such as cell size, number of directions, and block stride.
- d_h : It is the dimension of the HOG vector.

The HOG process is performed by normalizing the features and extracting them as a vector.

Obtaining ViT Attributes: The input image is also divided into fixed-size patches to be passed to ViT. Each patch $x_p \in \mathbb{R}^{p^2 \cdot C}$ is transformed into a higher-dimensional representation by a linear embedding layer as shown in Eq. (10).

$$z_0 = \left[x_{cls}; x_p^1 E; x_p^2 E; \dots; x_p^N E; \right] + PE \quad (10)$$

Here;

- x_{cls} : It is a classification indicator.
- $E \in \mathbb{R}^{(p^2 \cdot C) \times D}$: Embedding matrix
- $PE \in \mathbb{R}^{(N+1) \times D}$: Positional embedding
- N: Total number of patches
- D: Attribute size in transformer

Patches are passed through the self-attention mechanism, and classification information is extracted, especially over the x_{cls} vector. The obtained ViT feature is as seen in Eq. (11).

$$f_{ViT} = ViT(X) \in \mathbb{R}^{d_v} \quad (11)$$

4.1. Feature fusion

The main originality of the proposed “FeatFusion-ViT” model is that it extracts deep features extracted by ViT and HOG, which is a classical hand-crafted feature extraction method, in parallel ways and fuses them in a meaningful way. The aim at this stage is to make two separate feature spaces, which contain both local structural information and global contextual information, suitable for use in the same classification task.

Harmonization of Heterogeneous Feature Vectors: HOG and ViT attributes have different distributions and sizes. HOG features $f_{HOG} \in \mathbb{R}^{d_h}$ vector to represent low-level edges, directions, and patterns. ViT features (classification token output) contain high-level, global semantic information with the vector $f_{ViT} \in \mathbb{R}^{d_v}$. The two feature vectors are dimensionally unequal. Therefore, both vectors are passed through a separate learnable projection layer as in Eq. (12).

$$\tilde{f}_{ViT} = \sigma(W_v \cdot f_{ViT} + b_v), \tilde{f}_{HOG} = \sigma(W_h \cdot f_{HOG} + b_h) \quad (12)$$

Here;

- σ : ReLU activation function
- $W_v \in \mathbb{R}^{d_f \times d_v}, W_h \in \mathbb{R}^{d_f \times d_h}$: Learnable weights
- d_f : Common target size of both vectors

By applying this operation, two vectors at different scales are transformed into the feature space of the same dimension.

Feature Fusion Strategy: After the projection, the two vectors are combined as shown in Eq. (13). This study employs direct fusion due to its simplicity and efficiency.

$$f_{fused} = [\tilde{f}_{ViT} \parallel \tilde{f}_{HOG}] \in \mathbb{R}^{2d_f} \quad (13)$$

This combination vector is then optimized for classification by passing it through a learnable Multilayer Perceptron (MLP) layer as in Eq. (14).

$$z = \phi(W_{fusion} \cdot f_{fused} + b_{fusion}) \quad (14)$$

$$\hat{y} = Softmax(W_c \cdot z + b_c)$$

Here;

- ϕ : Activation function (ReLU or GELU)
- $W_c \in \mathbb{R}^{C \times d_z}$: Weights representing output classes
- C: Number of classes
- $\hat{y}$: Predicted class distribution

Table 2

Experimental results based on datasets.

Methods	PathMNIST		ChestMNIST		DermaMNIST	
	AUC	ACC	AUC	ACC	AUC	ACC
Ali et al. [12]		0.920		0.949		0.769
Manzari et al. [14]	0.999	0.977	0.823	0.967	0.950	0.817
Prakash et al. [8]	0.983	0.937			0.876	0.751
ResNet-18 (28) [9]	0.983	0.907	0.768	0.947	0.917	0.735
ResNet-18 (224)	0.989	0.909	0.773	0.947	0.920	0.754
ResNet-50 (28) [9]	0.990	0.911	0.769	0.947	0.913	0.735
ResNet-50 (224)	0.989	0.892	0.773	0.948	0.912	0.731
auto-sklearn	0.934	0.716	0.649	0.779	0.902	0.719
AutoKeras	0.959	0.834	0.742	0.937	0.915	0.749
Google AutoML Vision	0.944	0.728	0.778	0.948	0.914	0.768
Proposed method	0.998	0.981	0.843	0.974	0.953	0.859

The combination of these two approaches enables modeling of both global and local information levels. The projection layers applied before combining the outputs of these components make these information sources with different qualities comparable by bringing them to the same dimensional and semantic plane. In this way, the model is prevented from being overly dependent on a certain set of features during the learning process, and the risk of overfitting is reduced. Classification: The combined feature vector is passed through a fully connected layer to produce class probabilities as seen in Eq. (15).

$$\hat{y} = Softmax(W \cdot f_{fused} + b) \quad (15)$$

The pseudo code in Fig. 4 shows the basic process steps of the proposed method in detail. These steps include extracting global features via the vision transformer, obtaining local structural features via the Histogram of Oriented Gradients method, merging these two feature sets, and then performing the classification process via the multilayer perceptron (MLP). With this algorithmic representation, the step-by-step operation and modular structure of the method are clearly revealed.

The input image is processed in parallel by two separate branches: one is ViT-based global feature extraction, and the other is HOG-based hand-crafted feature extraction. The features obtained from these two branches are then combined and transferred to the projection layer (MLP) for classification. As seen in Fig. 5, it highlights that ViT and HOG features are complementary to each other and how these two different representations are combined into an integrated classification structure.

5. Experimental results

In this study, the proposed “FeatFusion-ViT” method offers a novel approach to medical image classification by integrating both traditional and deep learning-based feature extraction methods at the structural level. The approach’s success was investigated on two datasets to evaluate both its generalizability to different medical imaging tasks and its diagnostic performance on clinical-level mammograms. The MedMNIST dataset consists of numerous subsets encompassing different organs, tissues, and disease types, with varying class numbers, unbalanced data distributions, and single- to multi-channel image formats. This structure provides a suitable environment to evaluate the proposed method’s generalization ability on low-resolution and diverse medical images. In contrast, the CBIS-DDSM dataset consists of high-resolution mammograms and includes samples labeled as benign or malignant by radiologists. Thus, the model was tested on both low-resolution standardized images and high-resolution clinical data. Experimental results on subdatasets within the Medmnist cluster are shown in Tables 2–5, and experimental results on the CBIS-DDSM dataset are shown in Table 6.

Experimental results demonstrated that the HOG + ViT architecture provides superior performance in terms of accuracy and AUC-ROC

```

// Function: forward(image)
// Purpose: Fusing VIT and HOG features

function forward(image):

    STEP 1: vit_feat = vit(image) → Use Vision Transformer to capture the semantic and
    structural

    STEP 2: hog_feat = compute_hog(image) → Use HOG (Histogram of Oriented Gradients) to
    extract localized

    STEP 3: fused_feat = concatenate(vit_feat, hog_feat) → Concatenate both feature vectors into
    a unified representation

    STEP 4: fusion_output = MLP(fused_feat) → Project the combined features through an MLP

    STEP 5: output = classifier(fusion_output) → Make final prediction // Output class scores or
    probabilities

    STEP 6: Return the prediction result

```

Fig. 4. Pseudocode of the proposed method.

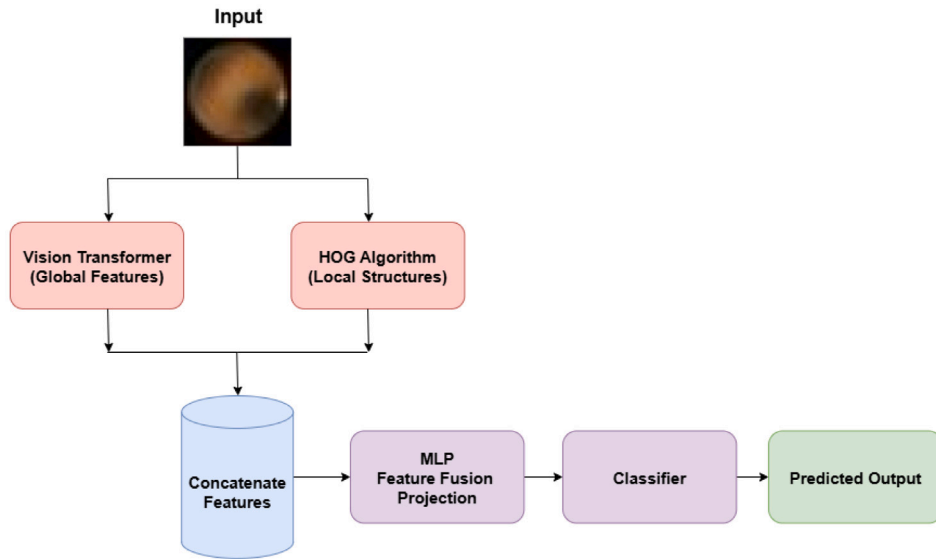


Fig. 5. Flow chart.

Table 3

Experimental results based on datasets.

Methods	OCTMNIST		Pne.MNIST		RetinaMNIST	
	AUC	ACC	AUC	ACC	AUC	ACC
Ali et al. [12]		0.800		0.944		0.561
Manzari et al. [14]	0.996	0.952	0.997	0.973	0.785	0.578
Prakash et al. [8]	0.935	0.843	0.965	0.926	0.763	0.565
ResNet-18 (28) [9]	0.943	0.743	0.944	0.854	0.717	0.524
ResNet-18 (224)	0.958	0.763	0.956	0.864	0.710	0.493
ResNet-50 (28) [9]	0.952	0.762	0.948	0.854	0.726	0.528
ResNet-50 (224)	0.958	0.776	0.962	0.884	0.716	0.511
auto-sklearn	0.887	0.601	0.942	0.855	0.690	0.515
AutoKeras	0.955	0.763	0.947	0.878	0.719	0.503
Google AutoML Vision	0.963	0.771	0.991	0.946	0.750	0.531
Proposed method	0.997	0.968	0.993	0.954	0.809	0.612

Table 4

Experimental results based on datasets.

Methods	BreastMNIST		BloodMNIST		TissueMNIST	
	AUC	ACC	AUC	ACC	AUC	ACC
Ali et al. [12]		0.911		0.967		0.760
Manzari et al. [14]	0.953	0.910	0.999	0.987	0.943	0.716
Prakash et al. [8]	0.901	0.923	0.997	0.974	0.961	0.700
ResNet-18 (28) [9]	0.901	0.863	0.998	0.958	0.930	0.676
ResNet-18 (224)	0.891	0.833	0.998	0.963	0.933	0.681
ResNet-50 (28) [9]	0.857	0.812	0.997	0.956	0.931	0.680
ResNet-50 (224)	0.866	0.842	0.997	0.950	0.932	0.680
auto-sklearn	0.836	0.803	0.984	0.878	0.828	0.532
AutoKeras	0.871	0.831	0.998	0.961	0.941	0.703
Google AutoML Vision	0.919	0.861	0.998	0.966	0.924	0.673
Proposed method	0.932	0.904	0.999	0.990	0.977	0.774

Table 5
Experimental results based on datasets.

Methods	OrganAMNIST		OrganCMNIST		OrganSMNIST	
	AUC	ACC	AUC	ACC	AUC	ACC
Ali et al. [12]		0.957		0.935		0.864
Manzari et al. [14]	0.999	0.973	0.999	0.961	0.987	0.851
Prakash et al. [8]	0.995	0.944	0.986	0.913	0.959	0.803
ResNet-18 (28) [9]	0.997	0.935	0.992	0.900	0.972	0.782
ResNet-18 (224)	0.998	0.951	0.994	0.920	0.974	0.778
ResNet-50 (28) [9]	0.997	0.935	0.992	0.905	0.972	0.770
ResNet-50 (224)	0.998	0.947	0.993	0.911	0.975	0.785
auto-sklearn	0.963	0.762	0.976	0.829	0.945	0.672
AutoKeras	0.994	0.905	0.990	0.879	0.974	0.813
Google AutoML Vision	0.990	0.886	0.988	0.877	0.964	0.749
Proposed method	0.997	0.980	0.999	0.966	0.989	0.865

Table 6
Experimental results on CBIS-DDSM dataset.

Methods	AUC	Specificity	Sensitivity	Accuracy
Sajid et al. [19]	–	–	–	0.915
Mannarsamy et al. [20]	–	0.952	0.947	0.992
Sharma et al. [21]	–	–	–	0.996
Vision Transformer	0.851	0.897	0.834	0.878
Proposed method	0.985	0.955	0.968	0.996

compared to ViT-only or CNN-based approaches, especially in cases where class discrimination is low or data imbalance is high. Moreover, the proposed method demonstrates superior overall classification performance by outperforming hybrid approaches combining hand-crafted and deep features, such as SIFT+CNN and HOG-LBP+CNN.

For deep learning models to be used confidently in clinical decision support systems, the interpretability of decision mechanisms is crucial. In this context, Grad-CAM is an effective explainable artificial intelligence (XAI) method that allows visualization of the image regions the model focuses on when making predictions [33]. Grad-CAM analysis is significant tool for verifying which information the model uses and how, especially in hybrid architectures that combine ViT-based complex attention mechanisms and HOG-enriched local features. A Grad-CAM analysis was performed on BloodMNIST and CBIS-DDSM datasets from MedMNIST datasets to evaluate the performance of the proposed “FeatFusionViT” architecture on decision results. The BloodMNIST results presented in Fig. 6 show that the model focuses heavily on central regions corresponding to the cell’s nucleus when performing inter-class discrimination. In the example of Class 3, the concentration of activations on nuclear boundaries and internal tissue suggests that the HOG component supports local tissue continuity while the ViT blocks process the global context. The CBIS-DDSM results presented in Fig. 7 reveal that the model develops different focusing strategies in benign and malignant samples. In the benign sample, “FeatFusionViT” assigns low-level activation to large, homogeneous tissue regions, while in the malignant sample, the local, high-density region where the lesion is located is clearly highlighted. This suggests that local structural distortions are captured by HOG-based features and supported by ViT in malignant classification.

Overall, both figures confirm that FeatFusionViT makes decisions by integrating both global contextual information and local structural details, providing an explainable and reliable decision mechanism by focusing on consistently meaningful regions across different datasets.

6. Discussion

In this study, we propose “FeatFusion-ViT”, a novel hybrid classification approach that integrates the global context modeling capabilities of the ViT architecture with HOG based local structural feature extraction. Experimental findings demonstrate that the proposed method

achieves statistically significant improvements over ViT-based models alone in terms of accuracy and AUC-ROC metrics on the MedMNIST and CBIS-DDSM datasets. The success of “FeatFusion-ViT” relies on the integration of the high-level global representations learned by ViT with HOG-based local edge and texture information. This multi-level representation structure allows the model to learn discriminative features more effectively, particularly on data subsets with high inter-class similarity or limited sample size. Indeed, the model has been observed to produce consistent and generalizable results on the high-resolution mammography images in the CBIS-DDSM dataset, as well as on datasets with unclear class boundaries, such as PathMNIST and OrganAMNIST.

Furthermore, training only the final layers of the ViT architecture and keeping the HOG features constant significantly reduces the risk of overfitting in small-scale datasets, reducing computational costs. In this respect, “FeatFusion-ViT” offers a balanced solution between computational efficiency and classification performance in systems with limited hardware resources. The model’s performance was compared with hand-crafted and deep learning-based hybrid methods in the literature, such as SIFT+CNN and HOG-LBP+CNN, and it was determined that “FeatFusion-ViT” achieved higher accuracy and AUC-ROC values compared to these approaches. However, the method has some limitations. Extracting HOG features with fixed parameters can prevent the adequate representation of specific details in images with complex textural structures. Furthermore, while initializing the ViT architecture with pre-trained weights improves overall performance, the difference in distribution between the pre-training data and the target area can limit classification performance, especially in small datasets.

In conclusion, the “FeatFusion-ViT” architecture combines traditional feature extraction techniques with modern transformation-based approaches to provide a robust and generalizable solution for highly complex datasets such as medical image classification. The proposed method achieves high accuracy and AUC-ROC results on both the MedMNIST subsets and the CBIS-DDSM dataset, demonstrating the effectiveness of the “FeatFusion-ViT” approach in clinical image analysis. Furthermore, Grad-CAM analyses demonstrate that the model focuses on meaningful and class-specific regions in both datasets when making decisions, confirming that “FeatFusion-ViT” can accurately integrate both global context and local structural details. These explainability results demonstrate that the architecture’s high performance is based on a consistent and interpretable decision mechanism, rather than a random one, and support the method’s potential for clinically reliable use.

CRedit authorship contribution statement

Muhammed Telçeken: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Formal analysis, Conceptualization.

Declaration of competing interest

The author declares that he has no known competing financial interests or personal relationships that could have appeared to influence the work reported in the manuscript.

Author have approved the final article and declare that the submission is original, has not been published previously, and is not under consideration for publication elsewhere.

Data availability

Data will be made available on request.

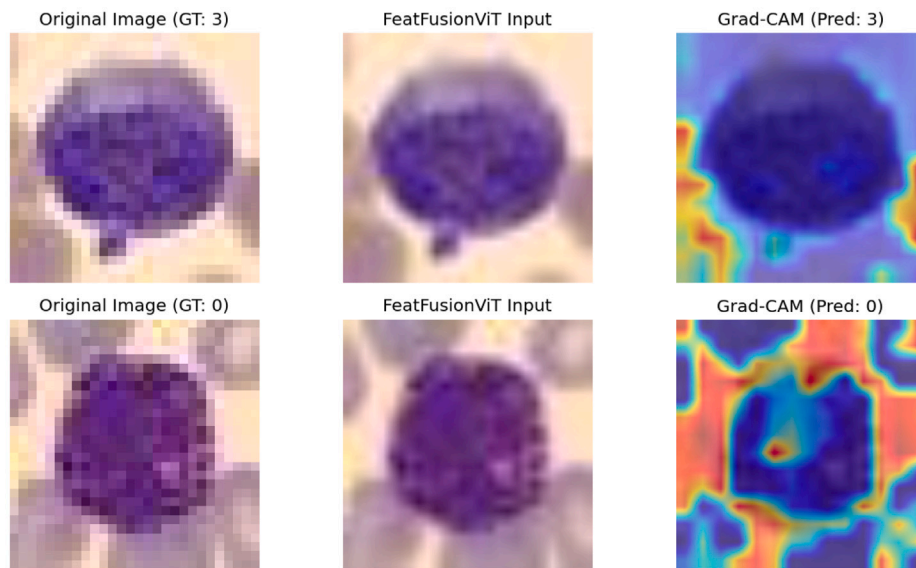


Fig. 6. Grad-CAM map showing FeatFusionViT focusing on BloodMNIST dataset.

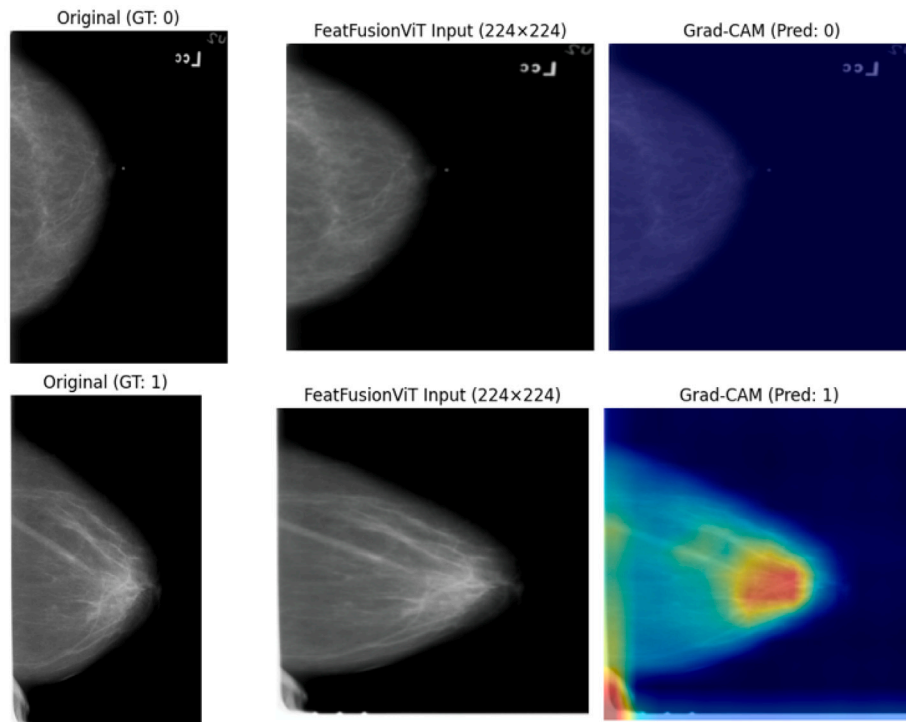


Fig. 7. Grad-CAM map showing FeatFusionViT focusing on CBIS-DDSM dataset.

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